In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix,ConfusionMatrixDisplay, clas
import warnings
warnings.filterwarnings('ignore')
%matplotlib inline
```

In [ ]:
```
df = pd.read_csv("Social_Network_Ads.csv")
```

In [ ]:
```
df
```

In [ ]:
```
df.shape
```

In [ ]:
```
df.info()
```

In [ ]:
```
df.describe()
```

In [ ]:
```
df.isna().sum()
```

In [ ]:
```
histplot = sns.histplot(df['Age'], kde=True, bins=10, color='red', alpha=
for i in histplot.containers:
     histplot.bar_label(i,)
plt.show()
```

In [ ]:
```
histplot = sns.histplot(df['EstimatedSalary'], kde=True, bins=10, color='
for i in histplot.containers:
     histplot.bar_label(i,)
plt.show()
```

In [ ]:
```
df["Gender"].value_counts()
```

In [ ]:
```
def gender_encoder(value):
  if (value == "Male"):
   return 1
  elif (value == "Female"):
   return 0
  else:
   return -1
```

In [ ]:
```
df["Gender"] = df["Gender"].apply(gender_encoder)
df
```

In [ ]:
```
df["Purchased"].value_counts()
```

In [ ]:
```
sns.pairplot(df,hue='Purchased',palette='bwr')
plt.show()
```

In [ ]:
```
sns.heatmap(df.corr(), annot=True)
plt.show()
```

In [ ]:
```
x = df[["User ID","Gender","Age", "EstimatedSalary"]]
y = df["Purchased"]
```

In [ ]:
```
scaler = StandardScaler()
x = scaler.fit_transform(x)
```

In [ ]:
```
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2,
```

In [ ]:
```
x_train.shape, x_test.shape, y_train.shape, y_test.shape
```

In [ ]:
```
model = LogisticRegression()
```

In [ ]:
```
model.fit(x_train, y_train)
```

In [ ]:
```
y_pred = model.predict(x_test)
y_pred
```

In [ ]:
```
model.score(x_train,y_train)
```

In [ ]:
```
model.score(x,y)
```

In [ ]:
```
cm = confusion_matrix(y_test, y_pred)
print(cm)
```

In [ ]:
```
disp=ConfusionMatrixDisplay(confusion_matrix=cm,display_labels=model.clas
disp.plot()
plt.show()
```

In [ ]:
```
print(f"TN value is {cm[0][0]}")
print(f"FP value is {cm[0][1]}")
print(f"FN value is {cm[1][0]}")
print(f"TP value is {cm[1][1]}")
```

In [ ]:
```
print(f"Accuracy score is {accuracy_score(y_test, y_pred)}")
print(f"Error rate is {1-accuracy_score(y_test, y_pred)}")
print(f"Precision score is {precision_score(y_test, y_pred)}")
print(f"Recall score is {recall_score(y_test, y_pred)}")
```

In [ ]:
```
print(classification_report(y_test, y_pred))
```

In [ ]:
```

```